

Nextcloud SaaS Application Support Lab

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Project Overview

This project simulates a SaaS Application Support environment using Nextcloud. It demonstrates both Tier 1 and Tier 2 support skills, including troubleshooting user login issues, file upload errors, missing files, quota problems, and file sharing with permissions.

Goals:

Project goals are to provide realistic SaaS support scenarios to add to my portfolio. As well as demonstrate command line and web based troubleshooting in Linux and Nextcloud environments. Lastly is to document support workflows, root causes, and resolutions.

Environment:

- **VMs:** Ubuntu 22.04 (Nextcloud server), Windows 11, Kali, pfSense (disabled)
- **Nextcloud version:** Latest stable
- **Web server:** Apache2
- **Database:** MariaDB
- **Networking:** VirtualBox NAT network for internet access

Project Structure

Scenario	Tier	Skills Highlighted
Password Reset / Account Recovery	Tier 1	User account management, password reset
User Cannot Sign In	Tier 2	Escalated troubleshooting, account lock/permission issues
Upload Fails Due to Quota	Tier 1–2	File system management, quota verification
File Uploaded but Missing / Reindex	Tier 2	Backend troubleshooting, <code>occ files:scan</code>
File Sharing / Permissions	Tier 2	File access control, restricted sharing, permissions

Scenario 1 — Password Reset / Account Recovery

Issue: User forgot password and cannot log in

Steps:

List all Nextcloud users

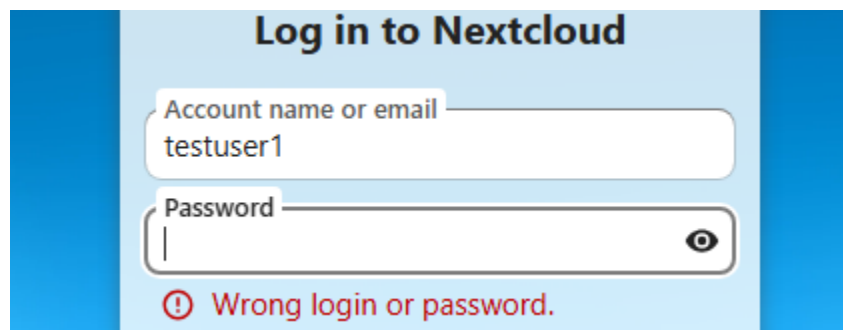
```
sudo -u www-data php /var/www/html/nextcloud/occ user:list
```

Reset password for testuser1

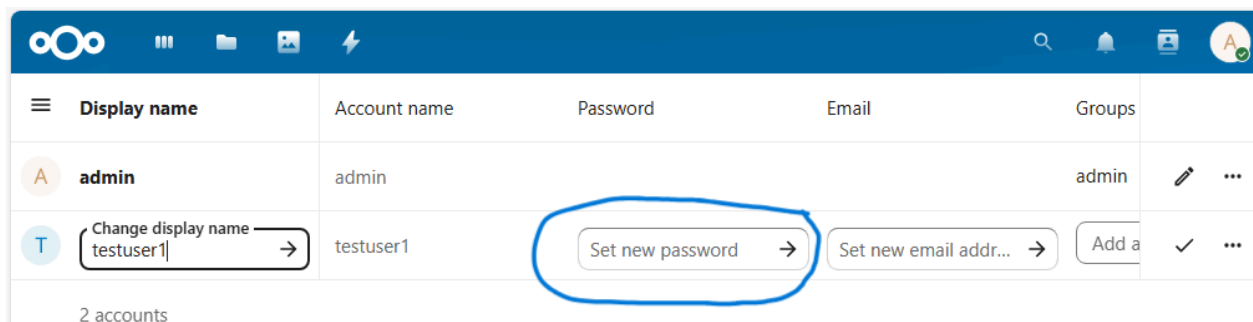
```
sudo -u www-data php /var/www/html/nextcloud/occ user:resetpassword testuser1
```

- Log in as admin → Users → Select **testuser1** → Reset password
- Confirm user can log in with new credentials

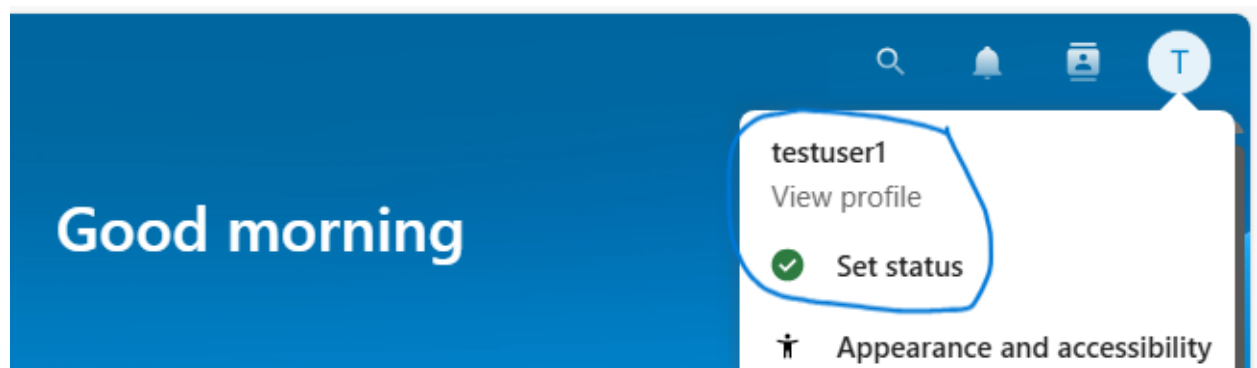
Login failure message:



Admin console with user selected at password reset screen:



Successful login:



Outcome: User successfully logged in, Tier 1 workflow completed

Scenario 2 — User Cannot Sign In

Steps:

1. Verify account is enabled:

```
sudo -u www-data php /var/www/html/nextcloud/occ user:info  
testuser1
```

2. Check permissions and groups, confirm no locks

3. Clear backend cache and repair:

```
sudo -u www-data php /var/www/html/nextcloud/occ  
maintenance:repair
```

Nextcloud logs showing failed login:

```
colem@Ubuntu:~$ sudo tail -n 30 /var/www/html/nextcloud/data/nextcloud.log  
[sudo] password for colem:  
{ "reqId": "f2TJZMhy7gVbXOH0L5ct", "level": 2, "time": "2026-01-13T14:53:59+00:00", "remoteAddr":  
"10.10.10.6", "user": "--", "app": "no app in context", "method": "POST", "url": "/nextcloud/ind  
ex.php/login", "message": "Login failed: admin (Remote IP: 10.10.10.6)", "userAgent": "Mozill  
a/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/143.0.0  
.0 Safari/537.36 Edg/143.0.0.0", "version": "32.0.3.2", "data": [] }  
{ "reqId": "x53oB1mwWZb2hp0xfQw9", "level": 2, "time": "2026-01-13T14:54:22+00:00", "remoteAddr":  
"10.10.10.6", "user": "--", "app": "no app in context", "method": "POST", "url": "/nextcloud/ind  
ex.php/login", "message": "Login failed: admin (Remote IP: 10.10.10.6)", "userAgent": "Mozill  
a/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/143.0.0  
.0 Safari/537.36 Edg/143.0.0.0", "version": "32.0.3.2", "data": [] }  
{ "reqId": "17LARuom5bvus3xsWETn", "level": 2, "time": "2026-01-14T14:12:06+00:00", "remoteAddr":  
"10.10.10.6", "user": "--", "app": "no app in context", "method": "POST", "url": "/nextcloud/ind  
ex.php/login", "message": "Login failed: testuser1 (Remote IP: 10.10.10.6)", "userAgent": "Mo  
zilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/143
```

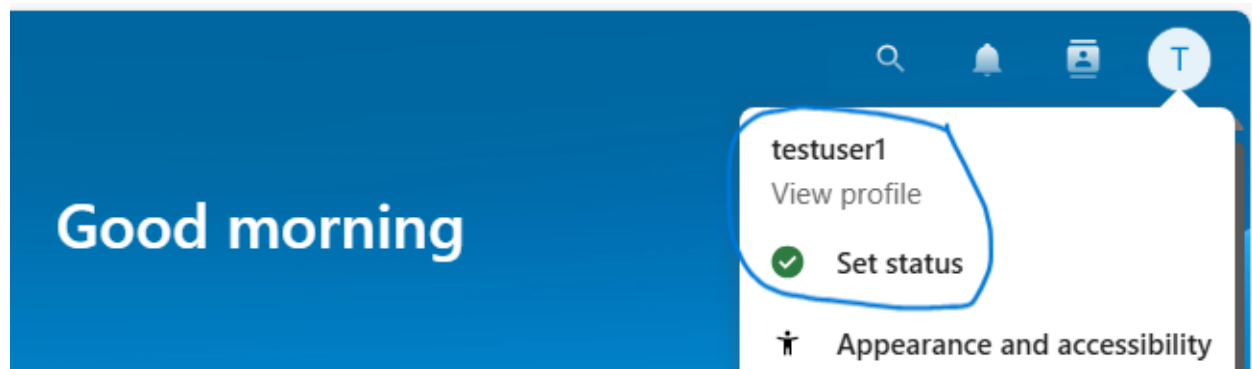
Verifying the account is enabled:

```
coleman@Ubuntu:~$ sudo -u www-data php /var/www/html/nextcloud/occ user:info testuser1
[sudo] password for coleman:
- user_id: testuser1
- display_name: testuser1
- email:
- cloud_id: testuser1@http://10.10.10.4/nextcloud
- enabled: true
- groups:
- quota: 1 GB
- storage:
  - free: 1015021169
  - used: 58720655
  - total: 1073741824
  - relative: 5.47
  - quota: 1073741824
- first_seen: 2026-01-14T14:42:40+00:00
- last_seen: 2026-01-14T17:59:02+00:00
- user_directory: /var/www/html/nextcloud/data/testuser1
- backend: Database
```

Maintenance:repair output:

```
coleman@Ubuntu:~$ sudo -u www-data php /var/www/html/nextcloud/occ maintenance:repair
- Repair MySQL collation
  - All tables already have the correct collation -> nothing to do
- Clean tags and favorites
  - 0 tags of deleted users have been removed.
  - 0 tags for delete files have been removed.
  - 0 tag entries for deleted tags have been removed.
  - 0 tags with no entries have been removed.
- Repair invalid shares
- Move .step file of updater to backup location
- Add move avatar background job
  - Repair step already executed
- Add preview cleanup background jobs
- Migrate oc_properties table to nextcloud schema
  - oc_properties table does not exist.
- Migrate oauth2_clients table to nextcloud schema
  - Update the oauth2_access_tokens table schema.
  - Update the oauth2_clients table schema.
  - Delete clients (and their related access tokens) with the redirect_uri starting with oc:// or ending with *
- Repair language codes
- Add log rotate job
- Clear frontend caches
  - Image cache cleared
  - JS cache cleared
- Clear every generated avatar
- Add preview background cleanup job
- Queue a one-time job to cleanup old backups of the updater
- Cleanup invalid photocache files for carddav
- Add background job to cleanup login flow v2 tokens
- Remove potentially over exposing share links
  - No need to remove link shares
```

Successful Login after Tier 2 troubleshooting:



Outcome: Issue resolved, demonstrating escalation workflow

Scenario 3 — Upload Fails Due to Quota

Issue: User cannot upload files due to storage limit

Steps:

Check quota for user

```
sudo -u www-data php /var/www/html/nextcloud/occ user:info  
testuser1
```

Increase quota if needed

```
sudo -u www-data php /var/www/html/nextcloud/occ user:modify  
testuser1 --quota 5G
```

- Retry file upload in browser
- Verify upload succeeds

Storage limit set:

A admin	Unlimited (54.8 MB used)
T testuser1	10 MB (54.8 MB used)

Upload file logs message error:

```
Windows NT 10.0; Win64; x64; AppleWebKit/537.36 (KHTML, like Gecko) Chrome/113.0.0.0 Safari/537.36 Edg/143.0.0.0", "version": "32.0.3.2", "exception": {"Exception": "OCA\\Text\\Exception\\UploadException", "Message": "No file uploaded or file size exceeds maximum of 2 MB", "Code": 0, "Trace": [{"file": "/var/www/html/nextcloud/apps/text/lib/Controller/AttachmentController.php", "line": 117, "function": "getUploadedFile", "class": "OCA\\Text\\Controller\\AttachmentController"}
```

Storage limit increase:

A admin	Unlimited (54.8 MB used)
T testuser1	1 GB (54.8 MB used)

Successful upload after storage increase:

Welcome to Nextcloud!

Here you can add a description or any other info relevant for the folder. It will show as a "Readme.md" and in the web interface also embedded nicely up at the top.

Files uploaded successfully

PDF

Nextcloud flyer.pdf

Last updated 42 minutes ago

Example.md

Last updated 42 minutes ago

Nextcloud intro.mp4

Last updated 42 minutes ago

Welcome to Nextcloud Hub...

Last updated 42 minutes ago

PDF

Nextcloud Manual.pdf

Last updated 42 minutes ago

Gorilla.jpg

Last updated 42 minutes ago

Scenario 4 — File Uploaded but Missing / Reindex

Issue: File uploaded but does not appear in UI

Steps:

```
# List files as www-data
```

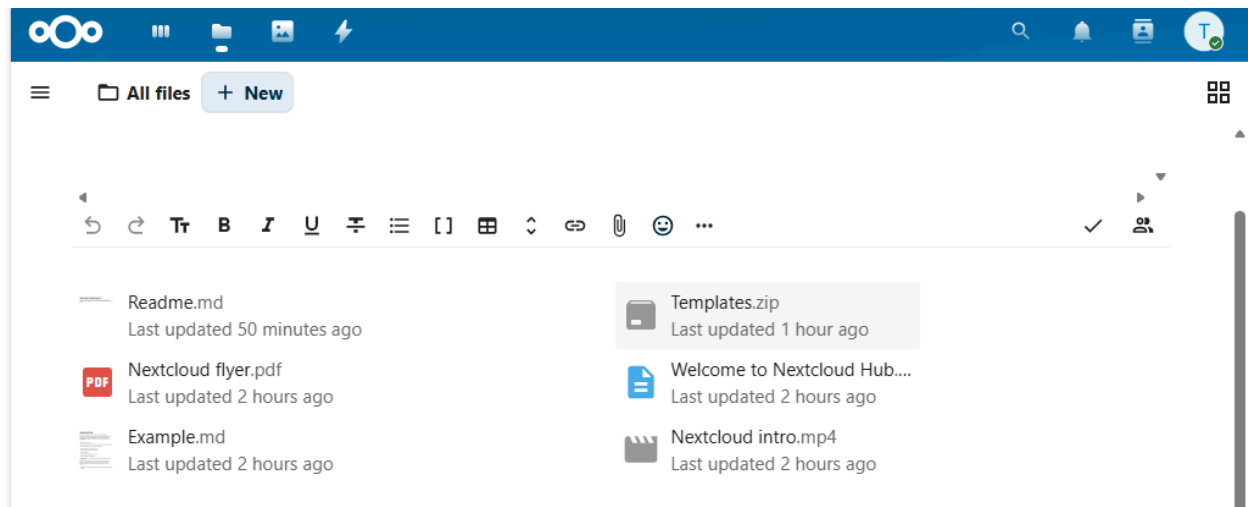
```
sudo -u www-data ls -lh  
/var/www/html/nextcloud/data/testuser1/files
```

```
# Reindex files for user
```

```
sudo -u www-data php /var/www/html/nextcloud/occ files:scan  
testuser1
```

- Refresh browser → file now visible

Browser showing missing file:



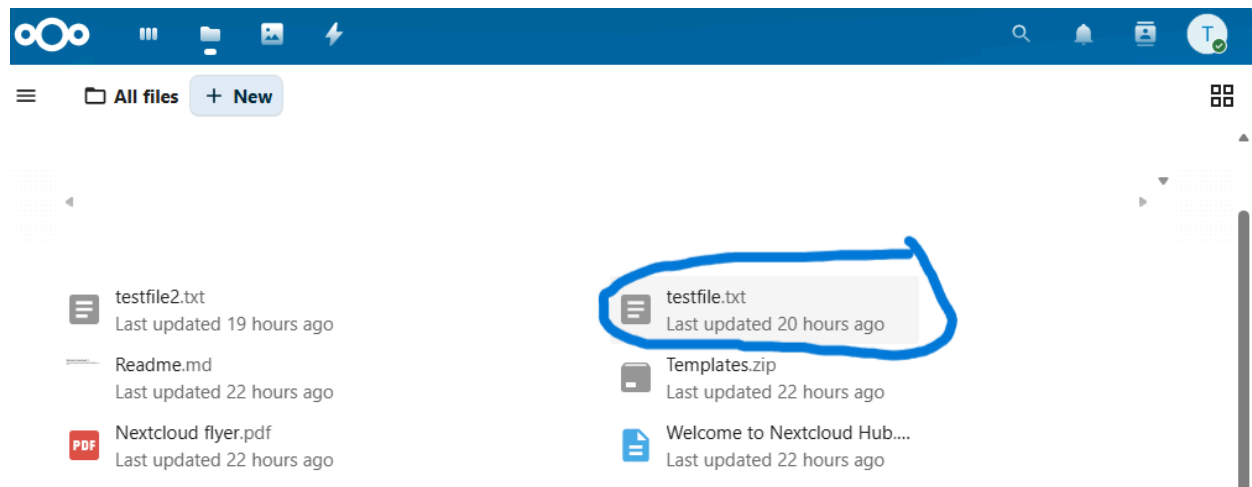
Terminal showing **ls -lh** output:

```
colemang@Ubuntu:~$ sudo -u www-data ls -lh /var/www/html/nextcloud/data/testuser1/files
[sudo] password for colemang:
total 28M
drwxr-xr-x 2 www-data www-data 4.0K Jan 14 09:42 Documents
-rw-r--r-- 1 www-data www-data 3.8M Jan 14 09:42 'Nextcloud intro.mp4'
-rw-r--r-- 1 www-data www-data 50K Jan 14 09:42 Nextcloud.png
drwxr-xr-x 2 www-data www-data 4.0K Jan 14 09:42 Photos
-rw-r--r-- 1 www-data www-data 197 Jan 14 10:24 Readme.md
-rw-r--r-- 1 www-data www-data 954K Jan 14 09:42 'Reasons to use Nextcloud.pdf'
drwxr-xr-x 2 www-data www-data 4.0K Jan 14 09:42 Templates
-rw-r--r-- 1 www-data www-data 3.1K Jan 14 09:42 'Templates credits.md'
-rw-r--r-- 1 www-data www-data 23M Jan 14 09:59 Templates.zip
-rw-r--r-- 1 www-data www-data 0 Jan 14 12:55 testfile2.txt
-rw-r--r-- 1 www-data www-data 0 Jan 14 12:06 testfile.txt
```

Terminal showing **occ files:scan** output:

```
colemang@Ubuntu:~$ sudo -u www-data php /var/www/html/nextcloud/occ files:scan testuser1
Starting scan for user 1 out of 1 (testuser1)
+-----+-----+-----+-----+-----+-----+-----+
| Folders | Files | New | Updated | Removed | Errors | Elapsed time |
+-----+-----+-----+-----+-----+-----+-----+
| 11      | 68    | 0   | 0       | 0       | 0       | 00:00:01     |
+-----+-----+-----+-----+-----+-----+-----+
```

Browser showing file after reindex:



Scenario 5 — File Sharing / Permissions

Issue: User wants to share a file with restrictions (view/edit/download)

Steps:

```
# Check files
```

```
sudo -u www-data ls -lh
```

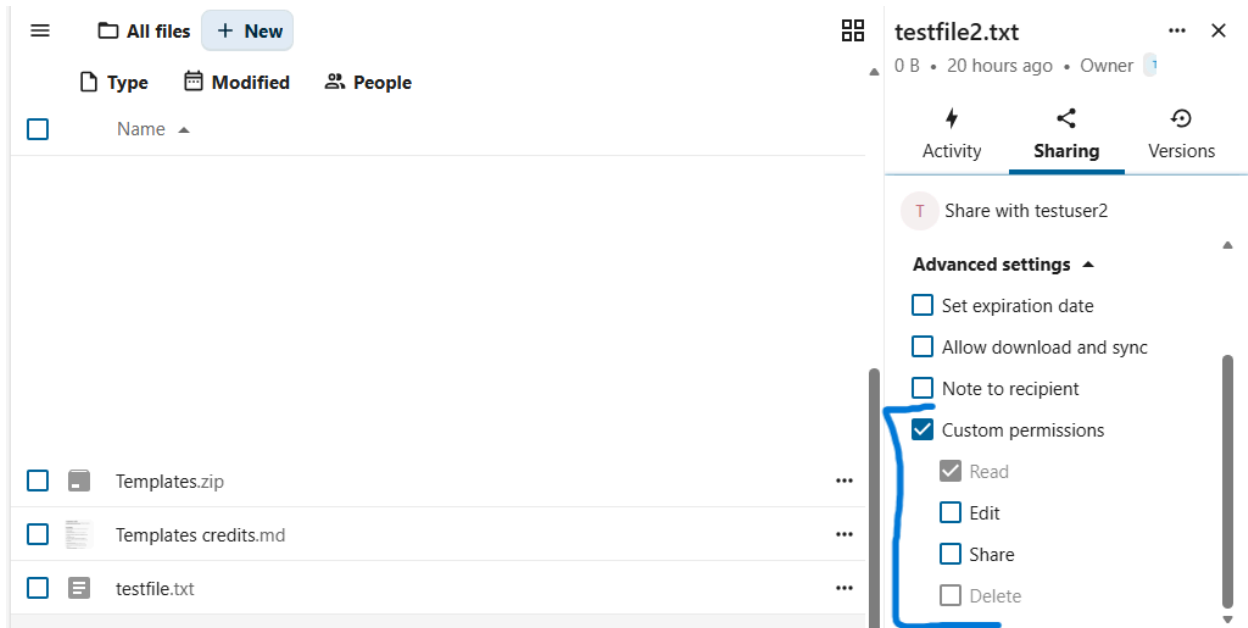
```
/var/www/html/nextcloud/data/testuser1/files
```

```
# Verify sharing app enabled
```

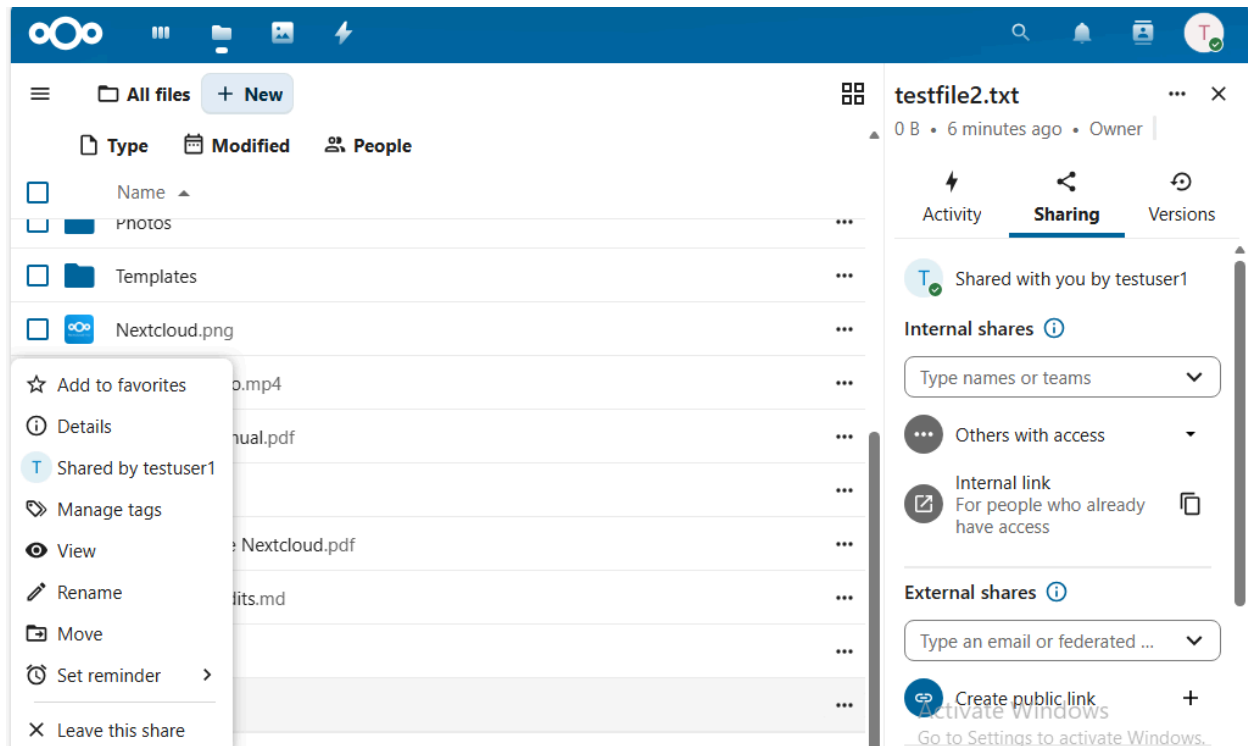
```
sudo -u www-data php /var/www/html/nextcloud/occ app:list | grep  
files_sharing
```

1. Log in as `testuser1` → Files → Select file → Share
2. Add `testuser2`
3. Configure permissions/restrictions:
 - Can view only
 - Disallow resharing
 - Disallow download
4. Log in as `testuser2` → confirm restricted options are hidden

Share dialog showing restricted options:



Recipient view with restricted actions not visible:



Commands Summary

Purpose	Command
List users	<code>sudo -u www-data php /var/www/html/nextcloud/occ user:list</code>
Reset password	<code>sudo -u www-data php /var/www/html/nextcloud/occ user:resetpassword <user></code>
Check files	<code>sudo -u www-data ls -lh /var/www/html/nextcloud/data/<user>/files</code>
Reindex files	<code>sudo -u www-data php /var/www/html/nextcloud/occ files:scan <user></code>
Check apps	<code>sudo -u www-data php /var/www/html/nextcloud/occ app:list</code>
Repair backend	<code>sudo -u www-data php /var/www/html/nextcloud/occ maintenance:repair</code>

Conclusion

This project demonstrated real-world SaaS application support workflows using Nextcloud, covering common Tier 1 and Tier 2 scenarios such as login issues, file upload failures, missing files, and permission-based sharing. Through structured troubleshooting and the use of both the web interface and command-line tools, each issue was successfully identified and resolved. Overall, this lab highlights practical SaaS support skills, clear documentation practices, and readiness for customer-facing technical support roles.